

A

Protein feature extraction and improvement



Position-specific scoring matrix X_P^{Evo}

$L \times 20$



SPOT-Contact-LM



$L \times L$



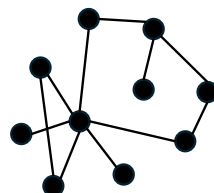
Contact probability matrix P



$L \times L$



Symmetric matrix M



Protein spatial network N_p



Graph attention auto-encoder



$L \times f$

Protein feature matrix $\hat{X}_P^{Evo}$

PTM site feature extraction and improvement

B

$\hat{X}_P^{Evo}(k)$

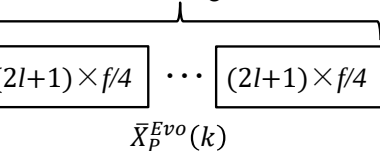
$(2l+1) \times 2f$



Multi-head self-attention



Concatenation of eight matrices



$\bar{X}_P^{Evo}(k)$

Residue-attention

$\hat{X}_P^{Evo}(k)$

W

$(2l+1) \times 2f$

$*$

$(2l+1) \times f$

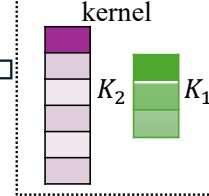
$\hat{X}_{P,K_1}^{Evo}(k)$

$\otimes$

$(2l+1) \times f$

$\hat{X}_{P,K_2}^{Evo}(k)$

1D convolution kernel



PTM site feature matrix $\hat{X}_P^{Evo}(k)$

$(2l+1) \times f$

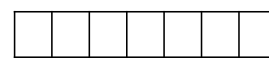
$\odot$

$\hat{X}_P^{Evo}(k)$

$(2l+1) \times 2f$

Attention pooling

$F_P^{Evo}(k)$



$\odot$ Convolution operator $*$ Column-wise Hadamard product

L Protein length

f Feature dimension defined in the last encoder layer of the GATE

l Number of residues in the upstream and downstream of the site

Focal loss

Adam optimizer

Mixup techniques

Fully connected layer

Sigmoid

ReLU

Hidden layer

Input layer

Prediction and optimization

C